

CSC4093/DSC4213: Deep Learning (2024/25)
Programming Assignment 02: Convolutional Networks and Transfer Learning
Due Date: 27/04/2026 | 11.59 pm

[This assignment carries 8% of the total grade.]

In this assignment, your task is to create a convolutional neural networks (CNNs) model for the classification of tea leaf disease images. The details of the dataset and the model are as follows.

Dataset:

The dataset includes images representing three different tea leaf diseases, with 100 images in each category.

Model:

VGG- or ResNet based CNN model built using transfer learning.

Sample Codes:

You can refer to the sample codes available on the LMS.

Languages and Frameworks:

You may use PyTorch or TensorFlow Keras libraries.

Deliverables:

1. Python scripts (.ipynb or .py) and screenshots of the outputs.
2. A report (PDF, maximum two pages) containing a performance comparison of the two models (e.g., accuracy and loss) using tables or plots.